

# Framing Risk Choice Teacher Notes

TOK Cognitive Science Activity Bank • Human Sciences • 20-30 min

|            |                |      |                     |
|------------|----------------|------|---------------------|
| Category   | Human Sciences | Time | 20-30 min           |
| Difficulty | Easy           | ID   | framing-risk-choice |

Big idea: The same evidence can lead to different choices when framed as gain or loss.

## Overview

Students make choices under gain and loss frames, then compare how wording changes apparently rational decisions.

## Student Prompt

Inspect Framing Risk Choice Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

## Concrete Stimulus Packet

|                                           |                                                                                                                                                                                                                                                 |                                                                                                                                             |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Framing Risk Choice Stimulus A</b>     | Student-facing stimulus 1: Gain frame: 200 of 600 people will be saved for sure, or a one-third chance all 600 are saved. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result. | Students inspect this exact card first and commit to a claim before hearing anyone else.                                                    |
| <b>Framing Risk Choice Stimulus B</b>     | Student-facing stimulus 2: Loss frame: 400 of 600 people will die for sure, or a one-third chance nobody dies. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.            | Use this for comparison, a second round, or a partner challenge.                                                                            |
| <b>Framing Risk Choice Reveal / Twist</b> | Student-facing stimulus 3: Loss frame: 400 of 600 people will die for sure, or a one-third chance nobody dies. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.            | Teacher reveal: connect the changed response to the big idea — The same evidence can lead to different choices when framed as gain or loss. |

## Actual Lesson Questions and Key

| # | Question for students                                                                | Teacher key / cue                                                                                             |
|---|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1 | After inspecting Framing Risk Choice Stimulus A, what is your first knowledge claim? | Teacher cue: look for a clear claim, not just a description.                                                  |
| 2 | Which exact detail from Framing Risk Choice Stimulus A did you use as evidence?      | Teacher cue: students should name visible words, data, source features, method features, or contextual clues. |
| 3 | What did you infer beyond the evidence?                                              | Teacher cue: this is where assumptions, framing, models, or perspective usually appear.                       |
| 4 | How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?                             | Teacher cue: confidence is data for the debrief, not a grade.                                                 |
| 5 | What missing information would most improve the knowledge claim?                     | Teacher cue: push for method, source, context, comparison, or definitions.                                    |
| 6 | Are decisions based on evidence, framing, or both?                                   | Teacher cue: students should move from the classroom example to a general TOK issue.                          |

## Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how the same evidence can lead to different choices when framed as gain or loss.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

## Setup Checklist

- Prepare mathematically equivalent gain-frame and loss-frame versions.
- Collect anonymous choices so students examine patterns, not individual judgement.

## Ready-to-Use Examples

- Gain frame: 200 of 600 people will be saved for sure, or a one-third chance all 600 are saved.
- Loss frame: 400 of 600 people will die for sure, or a one-third chance nobody dies.

## Suggested Timing

| Stage   | Teacher move                                                                       | Watch for                                                                   |
|---------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1       | Randomly give half the class a gain-framed version and half a loss-framed version. | Assumptions, confidence shifts, evidence claims, and language choices.      |
| 2       | Students choose between a certain option and a risky option.                       | Assumptions, confidence shifts, evidence claims, and language choices.      |
| 3       | Collect anonymous choices and compare class patterns by frame.                     | Assumptions, confidence shifts, evidence claims, and language choices.      |
| 4       | Reveal that the underlying outcomes were equivalent.                               | Assumptions, confidence shifts, evidence claims, and language choices.      |
| 5       | Debrief how framing can shape judgement without changing facts.                    | Assumptions, confidence shifts, evidence claims, and language choices.      |
| Debrief | Move from the activity result to a TOK knowledge question.                         | Students distinguishing method, evidence, interpretation, and overclaiming. |

## Teacher Language

- The numbers did not change; the frame did.
- If wording changes choices, what does that reveal about evidence and emotion?

## Common Misconceptions

- Students may treat class behaviour as universal human nature.
- Students may ignore ethical limits or context effects in human-subject inquiry.

## Assessment Look-Fors

- Connects behaviour to method, context, incentives, and ethical constraints.
- Avoids overgeneralising from one small sample.

## Differentiation and Adaptations

- Support: use smaller numbers and draw the outcomes.
- Challenge: include a third neutral-frame condition.

## **Extension Tasks**

- Students identify gain/loss framing in public health, climate, or school policy messages.
- Design a more transparent risk communication statement.

## **Related Activities**

- Anchoring Price Experiment
- Survey Wording Experiment
- Ultimatum Game